PolyAgora Phase-II

CTO Team Response & Engineering Deliverables

Convexity Exploitation Under Governance

PolygonEye AI Lab — CONFIDENTIAL, Internal Engineering

Baseline: V7.10 Strategy Sleeve Registry

Contents: CTO Response · Deliverable A · Deliverable B · Deliverable C

Draft v0.1

PolyAgora Phase-II — CTO Team Response

To: PolygonEye AI Lab — CTO / Quant Engineering / Systems Architecture **Re:** Phase-II Engineering Specification v1.0 (May 2026), "Convexity Exploitation Under Governance" **Status:** CTO counterproposal — Deliverables A, B, C + architectural challenge **Baseline:** V7.10 Strategy Sleeve Registry

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0. Executive Summary

This document responds to the Phase-II spec with the three requested deliverables (A — Runtime State Diagram, B — Recoverability Geometry Formalization, C — Convexity Participation Architecture) and an architectural counterproposal identifying the three highest-risk assumptions.

The single most important engineering finding precedes all three deliverables:

> **The "Governance Core" the spec describes is not a layer in the running system.** Its machinery — Reference Polygon, Survival Matrix `S_ij`, block/star logic, zone classification — is fully implemented, but it is *scoped inside one of four manifolds* (`v74d_q`). The actual top-level allocator (`v76α meta_blend`) is a rolling-Sharpe softmax with no `β`, no zone, and no survival matrix. The spec's instruction "the Governance Core must not be modified" therefore cannot be satisfied literally — the object it names does not exist as a modifiable layer.

CTO decision: the Governance Core will be **re-based on the V6.2 engine** (`polyagora_v62_engine.py`). This is the spec-literal reading: execution-graph Steps 2, 4, 5 and 6 describe V6.2 verbatim (the `(V,T,G,C,R)` coordinate, the 13×5 strategy×block admissibility, the survival matrix, the minimum-pairwise-`W` clustering). V6.2 becomes Layer 1; the V7.x stack (polygon, meta-blend, ADD-lite, rotation, sleeve registry) is re-wired to consume Layer 1's `β_t` / Zone / Governance-State outputs.

The re-base is the largest structural change in Phase-II and carries a regression risk addressed directly in the counterproposal (§4, Risk 1).

Document set. §§1–3 below are summaries. Each deliverable has a full companion document:

- `PolyAgora_PhaseII_Deliverable_A_State_Diagram.md` — 13-step graph, state schema, contracts
- `PolyAgora_PhaseII_Deliverable_B_Recoverability_Geometry.md` — 5 estimators, formulas, falsification protocol
- `PolyAgora_PhaseII_Deliverable_C_Convexity_Architecture.md` — 6-gate API, typed catalog, new candidates

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1. Deliverable A — Runtime State Diagram

1.1 Layer-1 reconciliation outcome

Layer 1 (Governance Core) is rebuilt from V6.2. It owns and emits, as **first-class top-level outputs**:

Output	V6.2 source	Definition
`X_t` — Reference Polygon coordinate	`compute_coordinate`	`(V,T,G,C,R) ∈ [-1,1]⁵`
`S_ij` — Survival Matrix	`rolling_survival_matrix`	continuous co-survival score `∈ [0,1]`
`A` — Admissibility matrix	Step 5 (see footnote 2)	13×5 strategy × block
`β_t` — β-Admissibility	`compute_x_sm_alignment` → governor	scalar `∈ [0,1]` recoverable-admissibility signal
Zone	`classify_block` + `star_zone`	5-mode classifier (see footnote 3)

Every other Phase-II layer (Convexity Sleeves, MRTP, Dynamic Kelly, Recoverability Geometry) operates strictly **downstream of, and conditional upon, these outputs**.

1.2 The 13-step runtime execution graph

Order is mandatory; each step's output feeds the next. Cadence: **W** = weekly, **E** = event-driven, **P** = persistent-state read.

#	Step	Target layer	Current code	Cadence
1	Market Data Ingestion	Runtime	`load_partner_xlsx` + market CSV	W
2	Polygon Projection → `(V,T,G,C,R)`	L1 Governance	`build_exogenous_x`, `compute_coordinate` (v62)	W
3	VAIDM Classification (6-axis → RUPTURE/HIGH_STRESS/ELEVATED/BENIGN)	L1 Governance	`Q_VAIDM` proxy only (v73) — see footnote 1	E
4	Survival Matrix Update `S_ij`	L1 Governance	`rolling_survival_matrix` (v62)	W
5	Admissibility Derivation (13×5)	L1 Governance	none — see footnote 2	W
6	Block / Zone Classification (5-mode)	L1 Governance	`classify_block`, `star_zone` (v62) — footnote 3	W
7	Recoverability Geometry Update	L5 Recoverability	none (ADD-lite `R`-component partial)	W / P
8	ADD Filtering (per-strategy decay)	L2 Sleeves / filter	`compute_add_lite_field`, `apply_asset_deformation` (v78)	W
9	MRTP Scoring	L3 MRTP	`compute_mrtp_score` (v76, dormant) — footnote 4	W
10	Dynamic Kelly Sizing `f_t`	L4 Kelly	v77 Kelly (haircut-only) — footnote 5	W
11	Sleeve Allocation	L2 Sleeves	`blend_sleeve` + registry (v710)	W
12	Contraction Map (Banach, `τ=0.60, λ=0.40`)	Runtime	contraction in v73/v74b — footnote 6	W

#	Step	Target layer	Current code	Cadence
13	Execution Output (allocation + validation + log)	Runtime	`evaluate` + CSV logging	W

1.3 State schema

Variable	Dim	Range	Update	Persistence	Normalization / smoothing
`X_t`	5	`[-1,1]`	weekly	recomputed	`tanh` per axis
`S_ij`	13×13	`[0,1]`	weekly	rolling window (126)	rolling corr + co-survival blend
`A`	13×5	`[0,1]`	weekly	recomputed	sigmoid
`β_t`	1	`[0,1]`	weekly	EWM-smoothed across steps	governor + alignment
Zone	categorical	5 modes	weekly	hysteresis on transitions	discrete
`R_t`	vector (v1)	`[0,1]`	weekly	history window	see Deliverable B
`C_t` corridor width	1	`[0,1]`	weekly	history window	see Deliverable B
`P_t` ridge pressure	1	`≥0`	weekly	history window	see Deliverable B
`F_t` fragmentation	1	`[0,1]`	weekly	history window	see Deliverable B
`f_t` Kelly multiplier	1	`[0, cap]`	weekly	hysteresis state persisted	see Deliverable C / §4
`Π_t` sleeve weights	n_sleeves	`[0,1]`	weekly	registry-persisted	zone-scaled

1.4 Storage contracts

- **Persisted across time steps:** `S_ij` rolling window, Recoverability history (Step 7), Kelly hysteresis state (Step 10), the Sleeve Registry, Zone-transition hysteresis buffer.
- **Recomputed fresh each bar:** `X_t`, `A`, `β_t`, MRTP scores, the final allocation vector.

Deliverable-A footnotes (spec/code reconciliation items)

1. **VAIDM is not implemented as a six-axis classifier.** Step 3 says "Load VAIDM report." In code, VAIDM exists only as `Q_VAIDM`, a single gross-exposure multiplier in `v73`. Deliverable A must pin VAIDM's provenance — external feed vs. computed from the macro panel — before Step 3 is buildable. *Open.*
2. **`derive_admissibility_from_survival()` does not exist.** Step 5 says the 13×5 admissibility is derived *from the survival matrix*. In code, admissibility is `σ(coef · polygon_coord)` — derived from the Reference Polygon, not `S_ij`. `S_ij` feeds block cohesion on a separate path. The rebased Layer 1 will implement Step 5 as specified; the polygon-derived path is retained as the admissibility *prior*.
3. **Zone classifier is 3-mode, spec wants 5.** V6.2 `star_zone` emits `zone1_star / zone2_transition / zone3_defensive`; `v75` emits 4 integer modes. The spec's 5 modes (`LOCAL_STAR / TRANSITION / LEAST_BAD / BUFFER / BOUNDARY`) require defining `LEAST_BAD` and `BUFFER`. There are also **three uncoordinated zone systems** today (v75 gross-gate

zones, v79 rotation zones, v710 sleeve-permission zones); Phase-II collapses these into the single Layer-1 5-mode classifier.

4. **MRTP formula differs.** `v76.compute_mrtp_score` implements `α·S + β·F − γ·D`; the spec's Step 9 / §3.3 wants `α·R + β·C − γ·F − δ·P`. The dormant code is a starting point, not the target — see Deliverable B for the `R/C/F/P` inputs.
5. **Kelly is inverted.** v77 Kelly is `f_t = mode_mult · f* · Φ` with `Φ ∈ [0,1]` and `kelly_cap = 1.0` — it can only *haircut* exposure. The spec's Dynamic Kelly must *expand* (see §4 Risk 3).
6. **Banach params differ.** Step 12 specifies `TAU_BUFFER=0.60, LAMBDA=0.40`; the contraction maps in `v73`/`v74b` use different values — to be standardized.
7. **Universe ambiguity.** V6.2's native universe is the 13 AGUR *strategies*; the V7.10 baseline metrics are on the 13 partner *futures*. The spec's "13×5" and "13-strategy universe" language matches V6.2. Deliverable A must pin which universe the rebased Layer 1 operates on, and the mapping between them.

2. Deliverable B — Recoverability Geometry Formalization

The deepest layer and the primary mathematical moat. Currently conceptual; Phase-II makes it a computable, runtime-updateable estimator set, tractable at weekly cadence over the 13-strategy universe across 2008–2026.

2.1 Candidate v1 estimators

Built from signals that already exist, to keep v1 falsifiable rather than speculative:

Geometry variable	v1 proxy estimator	Source
Corridor width `C_t`	count / dispersion of admissible blocks; spread of block membership scores	v62 `block_scores_from_x`, `A`
Ridge pressure `P_t`	rate-of-change of minimum pairwise `W_t`; contradiction-density buildup	v62 `rolling_survival_matrix`
Fragmentation field `F_t`	ADD-lite breadth (`B`) + synchronization (`S`) sub-components	v77/v78 ADD-lite field
Re-entry probability	conditional recovery frequency in a memory window after contraction	backtest-derived
Admissibility persistence	duration / stability of the current block's `β_t` above threshold	Layer-1 `β_t` series

2.2 Falsification protocol (the key methodological commitment)

Every estimator must, **before it is allowed to feed Kelly or MRTP**, pass a falsification test: it must measurably degrade through the 2008 and 2020 ruptures and recover through the post-2020 reflation. An estimator that does not move in the known episodes is rejected. This prevents "primary mathematical moat" from becoming unfalsifiable narrative.

2.3 Scalar vs. vector

`R_t` is kept a **vector** in v1 (per the spec's own Open Problem A, question 6). Collapse to a scalar only if a defensible aggregation survives §2.2.

2.4 Deliverable artifact

Backtested `R_t` (vector) over 2008–2026, annotated against known rupture (2008, 2020, 2022) and expansion (2009–2011, post-2020) episodes, with the §2.2 falsification result per estimator. Treated as a 6–8 week research track.

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3. Deliverable C — Convexity Participation Architecture

3.1 Formal sleeve admission pipeline — 6 gates

The current registry has 5 gates. Phase-II adds the sixth (Fragility Audit).

#	Gate	Metric	Status
1	Validation	finite standalone edge, ≥252 obs	exists
2	Deflated Sharpe	DSR noise floor (not the hard 0.95 AI-factory wall)	exists
3	Correlation	max corr to book **and to SPY** (hard ceiling `< 0.15`)	exists for book; **add SPY ceiling**
4	Walk-forward	OOS/IS degradation across ≥2 regimes	exists
5	Contribution	marginal Sharpe/Sortino — blending must not degrade the book	exists
6	**Fragility Audit**	stress test vs 2008 / 2020 / 2022 rupture windows	**new**

API: `evaluate_sleeve(spec, sleeve_returns, book_returns, cfg) → SleeveEvaluation`, extended with the SPY-correlation input and the Fragility Audit gate.

3.2 Typed sleeve catalog

Replace ad-hoc `SleeveSpec` strings with the spec's four types:

- **A — Crisis Trend.** Bond/USD/long-vol trend. *Status: bond-trend ADMITTED in V7.10.*
- **B — Expansion Convexity.** The Phase-II priority — the bull-market gap. *Status: no sleeve. New candidate required.*
- **C — Transition Convexity.** Cross-asset breakout, early momentum, low fragmentation. *Status: no sleeve. New candidate required.*
- **D — Rupture Asymmetry.** Tail-option / long-vol structure. *Status: data-blocked (no options/VIX-term-structure data) — flagged, not fabricated.*

3.3 New candidates within current data limits

The repo has only the 13-asset futures realized/forward PnL + a macro panel (VIX/SPY/HYG/TLT/GLD/CPER). Carry, Vol and Flow sleeves remain data-blocked.

- **Type B candidate — cross-sectional / relative-value expansion.** Long the strongest vs. short the weakest futures within the universe — captures persistent-expansion convexity *without* taking directional equity beta, the only way to also clear the Corr-SPY `< 0.15` gate (see §4 Risk 2).
- **Type C candidate — low-fragmentation cross-asset breakout.** Early-momentum entry gated by Layer-5 `F_t` so it only fires when fragmentation is low.

3.4 Decorrelation constraint matrix

A maximum-correlation matrix between every pair of admitted sleeves and between each sleeve and the main manifold, with the **SPY column as a hard wall** (`< 0.15`). Enforced both per-sleeve (gate 3) and at book level after each admission.

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4. Architectural Counterproposal — Three Highest-Risk Assumptions

The spec requests a structured challenge, not implementation agreement. The three highest-risk assumptions, with engineering mitigations:

Risk 1 — "The V7.10 baseline rests on a Governance Core that can be frozen."

The spec freezes the Governance Core and treats the V7.10 baseline (Sharpe 0.91, Max DD -6.46%) as resting on it. It does not. The baseline is produced by the **entire v75 → v76 → v78 → v79 stack**; the spec-described Core (V6.2) is not what runs, and V6.2 *alone* never produced those metrics. Re-basing Layer 1 onto V6.2 for spec-compliance therefore endangers the very baseline the spec wants protected.

Mitigation. Treat the re-base as a **behavior-bounded migration, not a rewrite**: (a) Deliverable A is the agreed contract before any module is built; (b) the rebased Layer 1 must reproduce the V7.10 baseline within a pre-agreed tolerance band as a hard acceptance gate — if it regresses, the re-base is rejected and we fall back to documenting the v75/v76 stack *as* the Core; (c) keep the current V7.10 stack runnable in parallel as the regression reference throughout Phase-II.

Risk 2 — "Expansion convexity can be captured while holding Corr SPY < 0.15."

The entire Phase-II thesis is harvesting persistent-bull / reflation upside. But Type B instruments (equity-trend acceleration, growth exposure) *are* equity beta. Capturing reflation convexity and holding near-zero SPY correlation pull in opposite directions — one of the Sharpe target or the correlation invariant will fail.

Mitigation. Define Type B as **cross-sectional / relative-value** expansion (long strong vs short weak within the futures universe) or commodity-curve expansion — not directional equity beta. Treat the acceptance criterion "narrow the gap vs Momentum by ≥30%" as the binding target, *not* full beta capture. Enforce SPY correlation as a hard registry gate both per-sleeve and at book level (§3.4).

Risk 3 — "Kelly leverage expansion is compatible with Max DD < 8%, given hysteresis."

Dynamic Kelly must *expand* gross toward (and possibly past) 1.0× during "stable expansion." But stable expansion is exactly the state preceding fast ruptures (2020 followed a calm bull). Hysteresis protects against transient ridge-pressure spikes — but it makes the system *slower to de-lever* into a genuine rupture from a high-leverage state. Leverage expansion and the drawdown floor are in direct tension.

Mitigation. Asymmetric Kelly dynamics: **slow expansion, instant contraction** — no hysteresis on the down-leg. Hard leverage cap (recommend **1.25×**, not uncapped). A non-negotiable cash / crisis-sleeve floor that survives even maximum Kelly expansion. Stress-test every Kelly schedule against a **synthetic instant-rupture-from-peak-leverage** scenario, not only the historical 2008/2020 paths.

Noted research risk (already flagged by the spec, not counted in the top three): Recoverability Geometry may not be cleanly computable from 13 return streams — mitigated by the falsification protocol in Deliverable B §2.2.

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5. Proposed Phasing

Track	Weeks	Scope
Deliverable A — state diagram + V6.2 re-base contract	1–2	gates everything
Deliverable C — 6-gate API + Type B/C candidate design	1–4	parallel to A
Deliverable B — recoverability estimators + 2008–2026 backtest	1–8	research track
Layer-1 re-base + regression gate (Risk 1 mitigation)	3–6	behavior-bounded
Layers 5 → 3 → 4 build (Recoverability → MRTP → Kelly)	6–12	dependent on B
Integration + stress test + CTO-target validation	10–14	acceptance

Acceptance: Sharpe > 1.2, Sortino > 1.5, Calmar > 1.0, Max DD < 8%, Corr SPY < 0.15, walk-forward OOS — *and* the Risk-1 regression gate (rebased Layer 1 reproduces the V7.10 baseline within tolerance).

6. Open Questions Returned to PolygonEye

1. VAIDM provenance — external six-axis feed, or computed from the macro panel?
2. Universe — does the rebased Layer 1 operate on the 13 AGUR strategies or the 13 partner futures? (Footnote 7.)
3. Definitions of the 5th and intermediate zone modes (`LEAST_BAD`, `BUFFER`).
4. Leverage cap — confirm 1.25× as the maximum admissible expansion.
5. Regression tolerance band for the Risk-1 acceptance gate.

Phase-II Deliverable A — Runtime State Diagram

Companion to: `PolyAgora_PhaseII_CTO_Response.md` (§1 is the summary; this is the full deliverable) **Status:** Draft v0.1 — the single source of truth for implementation sequencing **Scope:** the 13-step execution graph, state schema, storage contracts, layer contracts

> **Working assumptions** (flagged; pending PolygonEye confirmation — CTO_Response §6):
>
> - **A1 — Universe.** The rebased Layer 1 operates on the **13 partner futures** (BTC, CL, DX, ES, FESX, FGBL, GC, HG, NKD, SI, TN, ZS, ZW). V6.2's native AGUR-strategy universe is treated as the historical lineage; `v74` already re-targeted the block structure onto the futures. "13×5" = 13 futures × 5 blocks.
> - **A2 — VAIDM.** No external VAIDM feed exists. The six-axis VAIDM vector is **computed** from the macro panel (VIX/SPY/HYG/TLT/GLD/CPER) + realized PnL, generalizing the existing `Q_VAIDM` proxy.
> - **A3 — Zones.** The 5-mode classifier is defined in §3, Step 6.

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1. Purpose

Deliverable A fixes the runtime execution sequence, the state schema, and the storage contracts. Per the spec, it "must be agreed before any module is built." No layer may be implemented or modified independently of this document.

2. Layer-1 reconciliation (recap)

The Governance Core is **re-based on `polyagora_v62_engine.py`**. V6.2 becomes Layer 1 and emits, as first-class top-level outputs, the Reference Polygon `X_t`, the Survival Matrix `S_ij`, the 13×5 admissibility matrix `A`, the scalar β-Admissibility `β_t`, and the 5-mode Zone. Layers 2–5 operate strictly downstream.

3. The 13-step execution graph

Cadence key: **W** weekly · **E** event-driven · **P** persistent-state read.

Step 1 — Market Data Ingestion · Runtime · W

- **In:** partner XLSX (`realized_pnl`, `forward_pnl`), macro panel CSV.
- **Compute:** load, align to weekly index, normalize, compute returns + factor proxies.
- **Out:** `realized_pnl`, `forward_pnl`, macro panel `market`.
- **Code:** `load_partner_xlsx`, market CSV read. **No change.**

Step 2 — Polygon Projection · L1 Governance · W

- **In:** macro panel.

- **Compute:** `build_exogenous_x` → 5 raw features; `compute_coordinate` → `X_t = (V,T,G,C,R) ∈ [-1,1]⁵`. Feature lag = 1 (anti-hindsight).
- **Out:** `X_t`.
- **Code:** `build_exogenous_x`, `compute_coordinate` (v62). **No change.**

Step 3 — VAIDM Classification · L1 Governance · E

- **In:** macro panel + realized PnL (assumption A2).
- **Compute:** six-axis VAIDM vector; classify `RUPTURE / HIGH_STRESS / ELEVATED / BENIGN`; compute intensity multiplier `∈ [0,1]`.
- **Out:** `VAIDM_class`, `VAIDM_intensity`.
- **Code: new** — generalize `Q_VAIDM` (v73) into a six-axis classifier. *Footnote 1.*
- **Branch:** `RUPTURE` short-circuits Step 10 to Kelly FLOOR.

Step 4 — Survival Matrix Update · L1 Governance · W

- **In:** realized PnL over the trailing window (126).
- **Compute:** `S_ij = 0.40·corr_score + 0.40·(1−joint_bad) + 0.20·co_pos`, clipped `[0,1]`, unit diagonal.
- **Out:** `S_ij` (13×13).
- **Code:** `rolling_survival_matrix` (v62). **No change.**

Step 5 — Admissibility Derivation · L1 Governance · W

- **In:** `S_ij`, `X_t`, block geometry.
- **Compute:** the 13×5 admissibility matrix `A`. *Footnote 2 —* `derive_admissibility_from_survival()` does not exist; current admissibility is `σ(coef·X_t)` (polygon-derived). Phase-II builds Step 5 as: `A_polygon = σ(coef·X_t)` is retained as the **prior**; the survival matrix supplies a **cohesion modulation** `A_ij = A_polygon · g(block-internal S)`.
- **Out:** `A` (13×5).
- **Code: new** — `derive_admissibility_from_survival()`; reuses `_admissibility_v75` coefficients + `_block_internal_survival`.

Step 6 — Block / Zone Classification · L1 Governance · W

- **In:** `X_t`, `A`, `S_ij`.

- **Compute:** `classify_block` → active block; `star_zone` → strength + delta; micro-block clustering → minimum pairwise `W_t` per cluster. Map to the **5-mode** classifier (assumption A3):

Mode	Condition	Lineage
`LOCAL_STAR`	block strength ≥ z1 threshold, delta ≥ −0.08	v62 `zone1_star`
`TRANSITION`	strength ≥ z2 threshold, below z1	v62 `zone2_transition`
`LEAST_BAD`	no block strong; one block least-bad (min `W_t` still ≥ floor)	**new** — minimal deployment, not rupture
`BUFFER`	zone-transition cooldown / hysteresis holding state	**new** — anti-whipsaw dwell
`BOUNDARY`	VIX ≥ `boundary_vix` (35) — boundary block G active	v62 boundary

- **Out:** `Zone_t`, `block_t`, `β_t` (β-Admissibility, scalar — from X-SM alignment + governor).
- **Code:** `classify_block`, `star_zone` (v62) + **new** 5-mode mapping + `BUFFER` hysteresis. *Footnote 3.*

Step 7 — Recoverability Geometry Update · L5 · W/P

- **In:** `S_ij`, `A`, block geometry, `β_t` history, ADD-lite field.
- **Compute:** corridor width `C_t`, ridge pressure `P_t`, fragmentation `F_t`, recoverability `R_t` (re-entry probability + admissibility persistence). See **Deliverable B**.
- **Out:** `C_t, P_t, F_t, R_t`.
- **Code: new** — Layer 5.

Step 8 — ADD Filtering · L2 / governance filter · W

- **In:** base allocation, realized PnL.
- **Compute:** per-strategy Adaptive Decay Detector; redistribute weight among survivors within the admitted block.
- **Out:** decay-filtered allocation.
- **Code:** `compute_add_lite_field`, `apply_asset_deformation` (v78). **No change.**

Step 9 — MRTP Scoring · L3 · W

- **In:** `R_t, C_t, F_t, P_t` (Step 7), candidate trajectories.
- **Compute:** `MRTP_Score_i = α·R_i + β·C_i − γ·F_i − δ·P_i`; rank trajectories; decide expansion vs. contraction.
- **Out:** `MRTP_rank`, `convexity_decision`.
- **Code: new** — Layer 3. *Footnote 4* — `compute_mrtp_score` (v76) uses the old `α·S+β·F−γ·D` form; it is a starting point, not the target.

Step 10 — Dynamic Kelly Sizing · L4 · W

- **In:** `β_t, R_t, F_t, C_t`, `VAIDM_class`, prior Kelly state.
- **Compute:** `f_t = f(β_t, R_t, F_t, C_t)` with contraction hysteresis + anti-whipsaw; state ∈ {EXPANSION, CONTRACTION, FLOOR, RE-ENGAGEMENT}.
- **Out:** `f_t` (leverage multiplier, capped — assumption: 1.25×).
- **Code: new** — Layer 4, inverts v77 Kelly. *Footnote 5.*
- **Branch:** `VAIDM_class = RUPTURE` ⇒ Kelly FLOOR regardless of other inputs.

Step 11 — Sleeve Allocation · L2 · W

- **In:** `Zone_t`, `MRTP` output, admitted Sleeve Registry.
- **Compute:** sleeve type activations; zone-scaled weights within admission caps. See **Deliverable C**.
- **Out:** `Π_t` (sleeve weight vector).
- **Code:** `blend_sleeve`, `SleeveRegistry` (v710) — extended.

Step 12 — Contraction Map · Runtime · W

- **In:** prior allocation, target allocation.
- **Compute:** Banach contraction blend for block transitions, `τ_buffer=0.60, λ=0.40`. *Footnote 6* — standardize params vs v73/v74b.
- **Out:** smoothed allocation.

- **Code:** contraction maps in v73/v74b — standardized.

Step 13 — Execution Output · Runtime · W

- **In:** smoothed allocation × `f_t`.
- **Compute:** final allocation vector; validate vs envelope/constraints; log governance state, VAIDM reading, recoverability metrics.
- **Out:** `w_t`, run logs.
- **Code:** `evaluate` + CSV logging.

4. Data-flow diagram

```
[1 Ingest]
     │ realized/forward PnL, macro panel
     ▼
[2 Polygon] ──X_t──┐
     │              │
[3 VAIDM] ─────────┤  (E: RUPTURE branch ──────────────────┐
     │              │                                      │
[4 Survival S_ij]──┤                                       │
     │              │                                       │
[5 Admissibility A]┤                                       │
     │              │                                       │
[6 Block/Zone] ──β_t, Zone──┐                               │
     │                       │                               │
[7 Recoverability] ──R,C,F,P─┼────────────┐                  │
     │                       │            │                  │
[8 ADD Filter]               │            │                  │
     │                       ▼            ▼                  │
[9 MRTP Score] ◄────R,C,F,P──┘   [10 Kelly f_t] ◄─β,R,F,C────┤
     │ convexity_decision                │ (FLOOR if RUPTURE)◄┘
     ▼                                    │
[11 Sleeve Alloc Π_t] ◄──Zone, MRTP──────┘
     │
[12 Contraction Map]  (τ=0.60, λ=0.40)
     │
[13 Execution Output]  w_t = contracted_alloc × f_t
```

5. State schema

Symbol	Dim	Range	Producer	Consumers	Cadence	Norm / smoothing	Persistence
`X_t`	5	`[-1,1]`	S2	S5, S6	W	`tanh` per axis	recomputed
`VAIDM_class`	enum	4 modes	S3	S10	E	—	recomputed
`VAIDM_intensity`	1	`[0,1]`	S3	S7, S10	E	—	recomputed
`S_ij`	13×13	`[0,1]`	S4	S5, S6, S7	W	corr+co-survival	rolling 126

Symbol	Dim	Range	Producer	Consumers	Cadence	Norm / smoothing	Persistence
						blend	
`A`	13×5	`[0,1]`	S5	S6, S8, S11	W	sigmoid	recomputed
`β_t`	1	`[0,1]`	S6	S7, S9, S10	W	EWM (halflife ~10)	EWM state
`Zone_t`	enum	5 modes	S6	S11	W	`BUFFER` hysteresis	transition buffer
`C_t` corridor width	1	`[0,1]`	S7	S9, S10	W	see Deliverable B	history window
`P_t` ridge pressure	1	`≥0`	S7	S9	W	see Deliverable B	history window
`F_t` fragmentation	1	`[0,1]`	S7	S9, S10	W	see Deliverable B	history window
`R_t` recoverability	vector (v1)	`[0,1]`	S7	S9, S10	W	see Deliverable B	history window
`MRTP_rank`	n_traj	ordinal	S9	S11	W	—	recomputed
`f_t` Kelly mult	1	`[0, 1.25]`	S10	S13	W	hysteresis (asym.)	Kelly state
`Π_t` sleeve weights	n_sleeves	`[0,1]`	S11	S12	W	zone-scaled	registry
`w_t` final alloc	14	envelope	S13	—	W	contraction map	logged

6. Storage contracts

Persisted across time steps (carry state forward):

- `S_ij` rolling window — 126-bar realized-PnL buffer.
- Recoverability history (Step 7) — memory windows for re-entry probability and admissibility persistence.
- Kelly hysteresis state (Step 10) — current Kelly mode + dwell counters.
- Sleeve Registry — admitted set + per-sleeve metadata.
- Zone-transition hysteresis buffer (Step 6) — drives the `BUFFER` mode.
- `β_t` EWM smoothing state.

Recomputed fresh each bar (no carried state):

- `X_t`, `A`, MRTP scores, the final allocation vector.

7. Layer interaction contracts

- **L1 → all:** L1 promises `β_t ∈ [0,1]`, a valid 5-mode `Zone_t`, and `A` every bar. Downstream layers may **read** but never **write** L1 state.

- **L1 → L5:** L5 consumes `S_ij`, `A`, `β_t` history; L5 never feeds back into L1.
- **L5 → L3, L4:** `R/C/F/P` are inputs to MRTP and Kelly; they are the *only* path by which geometry reaches sizing.
- **L3 → L2:** MRTP may **modulate** sleeve activation within an admitted block; it may **not** override an L1 block selection (resolves spec Open Problem C, Q3).
- **L4 → Runtime:** Kelly produces a scalar `f_t`; it sizes, it does not select assets.
- **Rupture override:** `VAIDM_class = RUPTURE` forces Kelly FLOOR and caps sleeve activation to Type-A/D crisis sleeves only — a hard branch, not a soft modulation.

8. Footnotes (spec/code reconciliation — resolved within this deliverable)

1. **VAIDM** — built per assumption A2; six-axis classifier generalizing `Q_VAIDM`.
2. **`derive_admissibility_from_survival()`** — does not exist; built in Step 5 as polygon prior × survival-cohesion modulation.
3. **5-mode zones** — `LEAST_BAD` and `BUFFER` newly defined in Step 6; the three legacy zone systems (v75 gross-gate, v79 rotation, v710 sleeve-permission) collapse into this single Layer-1 classifier.
4. **MRTP formula** — `v76.compute_mrtp_score` uses `α·S+β·F−γ·D`; superseded by `α·R+β·C−γ·F−δ·P`.
5. **Kelly** — v77 Kelly is haircut-only (`cap=1.0`); Layer 4 rebuilds it to expand.
6. **Banach params** — Step 12 standardizes on `τ_buffer=0.60, λ=0.40`.
7. **Universe** — per assumption A1, the 13 partner futures.

Phase-II Deliverable B — Recoverability Geometry Formalization

Companion to: `PolyAgora_PhaseII_CTO_Response.md` (§2 is the summary; this is the full deliverable) **Status:** Draft v0.1 — research proposal, 6–8 week horizon **Scope:** formal estimators for the 5 geometry variables, tractable at weekly cadence over the 13-asset universe, 2008–2026

---

1. Purpose

Recoverability Geometry (Layer 5) is the deepest layer and the primary mathematical moat. The spec defines it conceptually; this deliverable makes it a **computable, runtime-updateable estimator set**. The governing question is not "how risky is the current position" but "**how many admissible futures does the current position preserve access to**."

The layer produces four runtime variables — `C_t` (corridor width), `P_t` (ridge pressure), `F_t` (fragmentation field), `R_t` (recoverability) — from five geometry estimators. `R_t` is the composite of the two estimators not already exposed as C/P/F (re-entry probability + admissibility persistence).

All estimators read realized data ≤ t−1 only (anti-hindsight).

2. The five geometry estimators

Notation: `S_ij` = Survival Matrix (Step 4); `p_b` = block membership scores from `block_scores_from_x` (b ∈ {A,B,C,D}); `A` = 13×5 admissibility; `β_t` = β-Admissibility; `r_t` = realized 13-asset return vector.

2.1 Corridor width `C_t`

"Volume of admissible trajectory space available from the current state." Wide when multiple blocks are jointly admissible **and** admissibility is high.

```
N_eff(t)  = exp( −Σ_b p_b ln p_b )           effective # of admissible blocks ∈ [1,4]
depth(t)  = Σ_b max(p_b − θ_C, 0)            admissibility mass above floor θ_C
C_t       = tanh( κ_C · (N_eff(t)/4) · depth(t) · β_t )
```

`C_t ∈ [0,1]`. Falls when the regime collapses onto one block (low `N_eff`) or when admissibility thins (`β_t` low). `θ_C ≈ 0.15`, `κ_C` calibrated so the 2009–2011 and post-2020 expansions sit in the top tercile.

2.2 Ridge pressure `P_t`

"Instability accumulation and contradiction density at the boundary of the corridor." Built from the **minimum pairwise survival** `W_min` — the spec's named cohesion metric.

```
W_min(t)  = min over admitted-block strategy pairs (i,j) of  S_ij(t)
level(t)  = 1 − W_min(t)                                     low cohesion = pressure
slope(t)  = max( 0, (W_min(t−k) − W_min(t)) / k )            cohesion eroding fast
P_t       = w_lvl · level(t) + w_slp · κ_P · slope(t)
```

`P_t ≥ 0`. Both terms non-negative; the slope term is the **early-warning** component (pressure *building* before the ridge becomes an active barrier). `k ≈ 4` weeks, `w_lvl = 0.5, w_slp = 0.5`.

2.3 Fragmentation field `F_t`

"Block incoherence, regime divergence, structural instability across the universe."

```
incoh(t)  = 1 − mean_offdiag( S_ij(t) )                      average pairwise incoherence
disp(t)   = z_clip( cross-sectional stdev of trailing r_t )  regime divergence
F_t       = 0.5 · incoh(t) + 0.5 · disp(t)
```

`F_t ∈ [0,1]`. Reuses the ADD-lite breadth/synchronization intuition but computes directly from `S_ij` so it is consistent with the rest of Layer 5. `z_clip` = z-score clipped to `[0,1]` via a logistic.

2.4 Re-entry probability

"Probability of re-engaging an admissible trajectory after contraction." Empirical conditional frequency over a memory window `M`.

```
A contraction episode opens when Zone enters {LEAST_BAD, BOUNDARY} or β_t < θ_β.
A re-entry occurs if Zone returns to {LOCAL_STAR, TRANSITION} within horizon H weeks.

ReEntry_t = (# re-entries within H) / (# contraction episodes)   over trailing M
```

`M ≈ 252 weeks` (rolling), `H ≈ 12 weeks`, `θ_β ≈ 0.35`. v1 uses the empirical frequency; a model-based form `σ(a + b·β_t + c·C_t − d·P_t)` is a v2 candidate once the estimators are validated.

2.5 Admissibility persistence

"Duration and stability of the current block's admissibility."

```
streak(t) = consecutive weeks with β_t > θ_β
stab(t)   = 1 − rolling_stdev( β_t, w_p )
Persist_t = 0.5 · tanh( streak(t) / τ_p ) + 0.5 · clip(stab(t), 0, 1)
```

`τ_p ≈ 26 weeks`, `w_p ≈ 13 weeks`. Distinguishes a stable equilibrium from a transient admissibility spike.

2.6 Composite recoverability `R_t`

```
R_t = 0.5 · ReEntry_t + 0.5 · Persist_t           (v1 scalar)
R_t = [ ReEntry_t , Persist_t ]                   (v1 vector — preferred, see §4)
```

3. Normalization, smoothing, memory windows

Estimator	Output range	Memory window	Smoothing
`C_t`	`[0,1]`	none (instantaneous)	EWM halflife 4w
`P_t`	`≥0` (clip 0–1 for MRTP)	4w slope window	EWM halflife 3w
`F_t`	`[0,1]`	trailing 13w for `disp`	EWM halflife 4w
ReEntry	`[0,1]`	252w rolling	none (already an average)
Persist	`[0,1]`	13–26w	none

Smoothing halflives are deliberately short (3–4w) — Layer 5 is faster than the Governance Core but slower than execution, preserving the spec's layer-separation principle.

4. Scalar vs. vector

`R_t` is kept a **vector** `[ReEntry_t, Persist_t]` in v1, per the spec's Open Problem A, question 6. Re-entry probability and admissibility persistence answer different questions (can I get back vs. is the current state stable) and may diverge — e.g. a long stable streak with low historical re-entry frequency. Collapsing to a scalar is deferred until the §5 falsification test shows the two move together; if they do not, the vector form is retained and MRTP/Kelly consume both components.

5. Falsification protocol — the key methodological commitment

Every estimator must pass a falsification test **before it is wired into MRTP (Step 9) or Kelly (Step 10)**. An estimator that does not respond to the known episodes is rejected, not tuned.

Required behavior over 2008–2026:

Episode	Window	`C_t`	`P_t`	`F_t`	`R_t`
GFC rupture	2008-04 → 2009-03	↓ low	↑ high	↑ high	↓ low
GFC recovery	2009-04 → 2011-12	↑ rising	↓ falling	↓ falling	↑ rising
QE bull	2012 → 2019	high, stable	low	low	high
COVID rupture	2020-02 → 2020-12	↓ sharp	↑ sharp	↑ sharp	↓ sharp
Post-2020 reflation	2021	↑ rising	↓	↓	↑
2022 rate shock	2022	↓ moderate	↑ moderate	↑ moderate	↓ moderate

Pass criterion: the estimator's episode-mean must sit in the correct tercile (top / mid / bottom) for ≥5 of the 6 episodes, and the COVID + GFC ruptures must both register as bottom-tercile `C_t`/`R_t` and top-tercile `P_t`/`F_t`. An estimator failing this is returned to formalization, not calibrated to pass.

`VAIDM` coupling (spec Open Problem A, Q5): under `VAIDM_class = RUPTURE`, recoverability must degrade — operationalized as a hard check that `R_t` mean over RUPTURE-classified bars < `R_t` mean over BENIGN bars by a margin (target ≥ 0.3).

6. Computational tractability

At weekly cadence over 13 assets, 2008–2026 ≈ 940 weekly bars:

- `S_ij` is 13×13 — all estimators are `O(13²)` per bar, trivially tractable.
- `N_eff`, `W_min`, `incoh` are single passes over `S_ij` / `p_b`.
- Re-entry probability is a rolling count — `O(M)` per bar, `M ≈ 252`.
- No optimization, no matrix inversion, no iterative solver. Full-history backtest runs in seconds. Tractability is **not** a constraint at this universe size.

7. Integration points

Estimator	Feeds	Role
`C_t` corridor width	MRTP (`+β·C`), Kelly	expansion confidence
`P_t` ridge pressure	MRTP (`−δ·P`)	contraction trigger
`F_t` fragmentation	MRTP (`−γ·F`), Kelly	contraction trigger
`R_t` recoverability	MRTP (`+α·R`), Kelly	primary expansion qualifier

8. Deliverable artifact

The completed Deliverable B ships: (1) the five estimator implementations; (2) the `R_t` vector backtested over 2008–2026 with the §5 episode annotations; (3) the falsification-test result per estimator (pass/fail + episode terciles); (4) calibrated constants (`κ_C, θ_C, k, τ_p, M, H, θ_β`). Treated as a 6–8 week research track running parallel to Deliverables A and C.

9. Open items

- Calibration of `κ_C` and the MRTP/Kelly coefficient weights is empirical — depends on Deliverable A's universe decision (assumption A1).
- The model-based re-entry form (§2.4 v2) is deferred pending v1 validation.
- Whether `F_t` and `P_t` are sufficiently independent to both feed MRTP, or whether one is redundant, is itself a §5 output.

Phase-II Deliverable C — Convexity Participation Architecture

Companion to: `PolyAgora_PhaseII_CTO_Response.md` (§3 is the summary; this is the full deliverable) **Status:** Draft v0.1 **Scope:** the formal 6-gate sleeve admission pipeline, the typed sleeve catalog, two new sleeve candidates, the decorrelation constraint matrix

---

1. Purpose

Convexity Sleeves (Layer 2) replace passive defensiveness with **governed convex participation**. The Phase-II gap is bull/reflation under-participation; the sleeve framework is the vehicle for closing it without destabilizing survivability. This deliverable formalizes how a candidate sleeve is admitted, how large it may grow as a function of governance state, and how sleeves are kept mutually decorrelated.

> **Data constraint.** The repo has only the 13-asset futures realized/forward PnL + the macro panel (VIX/SPY/HYG/TLT/GLD/CPER). Sleeves needing options, VIX term structure, or positioning data (Carry, Vol, Flow, Type-D tail) are **data-blocked** — flagged in §6, not fabricated.

2. The 6-gate admission pipeline

A candidate never enters the book on Sharpe alone. It clears an **ordered** gate pipeline; failing any single gate ⇒ `REJECTED` with a documented reason. The current registry (`polyagora_sleeve_registry.py`) implements gates 1–5; Phase-II adds gate 6.

2.1 Input schema

```
SleeveCandidate:
  spec            : SleeveSpec    # strategy_id, sleeve_type ∈ {A,B,C,D}, horizon,
                                  # economic_mechanism, regime_affinity, failure_modes,
                                  # polyagora_block, zone_permission
  sleeve_returns  : Series        # net of realistic transaction cost (cost_adjust)
  book_returns    : Series        # the current admitted book (v79 / v7.10)
  spy_returns     : Series        # for the SPY-correlation gate (new)
  weights_panel   : DataFrame     # for the Fragility Audit window slicing (new)
```

2.2 The gates

#	Gate	Metric	Threshold	Rejection reason
1	Validation	standalone Sharpe, n_obs	Sharpe ≥ 0.20, n ≥ 252	no real edge / too short
2	Deflated Sharpe	DSR p-value	`> dsr_floor` (0.50, noise floor — **not** the 0.95 AI-factory wall)	edge not separable from best-of-N luck

#	Gate	Metric	Threshold	Rejection reason
3	Correlation	`	corr	` to book **and** to SPY
4	Walk-forward	OOS/IS Sharpe degradation	ratio ∈ `[0.3, 1.3]` ("healthy"/"suspicious")	fragile out-of-sample
5	Contribution	blended vs book Sharpe & Sortino	blend ≥ book on both	blending degrades the book
6	**Fragility Audit** (new)	per-window Sharpe + Max DD over 2008 / 2020 / 2022	no rupture window worse than `−8%` DD contribution to the blended book	sleeve amplifies a known rupture

2.3 Contribution scoring formula

The registry weight a sleeve earns is diversification-scaled; the contribution score quantifies marginal benefit:

```
diversification = max(0, 1 − |corr_to_book|)
base_weight     = cap · diversification                  cap = 0.25 for a first sleeve
blend           = (1 − base_weight)·book + base_weight·sleeve
ContributionScore = (Sharpe(blend) − Sharpe(book)) / |Sharpe(book)|
                  + (Sortino(blend) − Sortino(book)) / |Sortino(book)|
```

Gate 5 passes iff `ContributionScore ≥ 0` on both terms. Among admitted sleeves the book share is allocated in descending `ContributionScore`.

2.4 Output

```
SleeveEvaluation: spec, metrics, dsr, degradation, corr_to_book, corr_to_spy,
                  fragility_audit, contribution, gates{6 bools}, admitted, base_weight, notes
```

3. Typed sleeve catalog

The ad-hoc `SleeveSpec.sleeve_type` string is replaced by the spec's four formal types, each with a regime target, a default zone-permission table, and an admission-cap profile.

Type	Regime target	Instruments (this universe)	Status
A — Crisis Trend	panic persistence, flight-to-quality, rate shock	bond/USD time-series trend	**ADMITTED** (bond-trend, V7.10)
B — Expansion Convexity	persistent bull acceleration, stable reflation	cross-sectional / curve expansion	**candidate** (§4.1)
C — Transition Convexity	emerging regime transitions, early breakout	low-fragmentation cross-asset breakout	**candidate** (§4.2)
D — Rupture Asymmetry	non-linear breakdowns, correlation collapse	tail-option / long-vol structure	**data-blocked** (§6)

4. New sleeve candidates

4.1 Type B — Cross-Sectional Expansion sleeve

The Phase-II priority. **Critically, it is *not* directional equity beta** — that is the only construction that can also clear the SPY-correlation gate (CTO_Response §4, Risk 2).

- **Signal.** Rank the 13 futures by trailing 6–12 month risk-adjusted return; long the top quartile, short the bottom quartile; normalize to gross 1; long/short, so market-direction-neutral by construction.
- **Economic mechanism.** Persistent-expansion regimes show *cross-sectional dispersion* — the leaders keep leading. The sleeve harvests that dispersion convexity without taking the index.
- **Regime affinity.** `LOCAL_STAR`, persistent reflation, widening corridor (`C_t` high).
- **Failure modes.** Sharp cross-sectional reversals; correlation-collapse ruptures where all assets move together (dispersion vanishes).
- **Expected behavior.** Positive marginal Sharpe in 2009–2011 and post-2020; near-zero SPY correlation by the long/short construction; gate 3 SPY `< 0.15` should pass.
- **Admission risk.** Gate 6 Fragility Audit — must not amplify 2020; expected to need zone-permission `0` in `BOUNDARY` / `RUPTURE`.

4.2 Type C — Low-Fragmentation Breakout sleeve

- **Signal.** Cross-asset breakout entry (price clears an N-week range) **gated by Layer-5 `F_t`** — the sleeve only fires when fragmentation is low, i.e. the breakout is structurally coherent rather than a fragmented false move.
- **Economic mechanism.** Early-transition persistence: a breakout into a low-`F_t` regime tends to continue; into a high-`F_t` regime it whipsaws.
- **Regime affinity.** `TRANSITION` zone, rising `C_t`, low `F_t`.
- **Failure modes.** High-fragmentation false breakouts (suppressed by the `F_t` gate); rangebound chop.
- **Expected behavior.** Contributes in regime-transition windows; the `F_t` gate is itself the decorrelation mechanism vs. the Type-A crisis-trend sleeve.

5. Maximum admissible sleeve allocation by governance state

The sleeve cap is a function of `Zone_t` and sleeve type — convexity sleeves expand in benign regimes and switch off in rupture; crisis sleeves do the opposite.

Zone	Type A (Crisis)	Type B (Expansion)	Type C (Transition)	Type D (Rupture)
`LOCAL_STAR`	0.10	**0.25**	0.10	0.00
`TRANSITION`	0.15	0.12	**0.25**	0.05
`LEAST_BAD`	0.20	0.05	0.10	0.10
`BUFFER`	hold prior	hold prior	hold prior	hold prior
`BOUNDARY` / RUPTURE	**0.25**	0.00	0.00	**0.25**

Caps are the *maximum* book share; the actual weight is `base_weight` (from §2.3) scaled by the zone cap and further by MRTP's convexity decision. `BUFFER` freezes sleeve weights at their prior value — the anti-whipsaw dwell state.

6. Dynamic intra-regime weight adjustment

Within an admitted zone, sleeve weights adjust with recoverability geometry, not just on zone changes:

```
Π_sleeve(t) = base_weight · zone_cap(Zone_t, type)
            · g_type( C_t, R_t, F_t )
```

- **Type B:** `g_B` rises with `C_t` and `R_t` (wider corridor, higher recoverability ⇒ more expansion convexity), falls with `F_t`.
- **Type C:** `g_C` peaks at intermediate `F_t` and rising `C_t`.
- **Type A:** `g_A` rises with `F_t` and `P_t` (crisis sleeve scales *into* stress).

This is the mechanism the spec's Open Problem B, Q4 asks for — sleeve weights track the geometry continuously, not only at discrete zone transitions.

7. Decorrelation constraint matrix

Maximum allowed correlations, enforced at gate 3 per-sleeve and re-checked at book level after each admission:

Pair	Max `|corr|`
sleeve ↔ sleeve	0.50
sleeve ↔ main manifold (v79 book)	0.70
sleeve ↔ SPY	**0.15** (hard wall — the inviolable invariant)
book-aggregate ↔ SPY	0.15

A candidate that would push the book-aggregate SPY correlation above 0.15 is rejected even if it individually passes gate 3.

8. Expected Sharpe improvement & stress-test framing

Per the spec's Deliverable-C ask, each admitted sleeve ships with: (1) a quantified marginal-Sharpe estimate from §2.3's `ContributionScore`; (2) the gate-6 Fragility Audit table — per-window Sharpe and Max DD over 2008 / 2020 / 2022. The bond-trend (Type A) reference: corr-to-book ≈ 0.28, flips the 2022 window from −0.84 to +0.44 Sharpe. The Type B sleeve's target is the Phase-II acceptance criterion — **narrow the underexposure gap vs. Momentum in post-2020 reflation by ≥30%** — not full beta capture.

9. Data-blocked sleeves (flagged, not built)

- **Type D — Rupture Asymmetry:** needs options / long-vol instruments — no data.
- **Carry:** needs futures term structure — no data.
- **Vol / Flow:** needs VIX term structure / dealer-gamma / CTA positioning — no data.

These remain registry placeholders with documented data requirements; they are not fabricated from proxies.

10. Open items

- The Type B lookback (6 vs 12 months) and quartile width are calibration choices — resolved against the gate pipeline, not against returns.
- The `g_type` functional forms (§6) depend on Deliverable B's validated `C/R/F` estimators.
- Whether the V7.10 bond-trend sleeve's flat zone-permission table is retained or migrated to the §5 cap profile is a registry-migration decision for Deliverable A.